

JOHN GWEN ISAAC D. AUSTRIA

Obando, Bulacan | 09484485458 | johngwenisaacaustria@gmail.com | [Portfolio](#)

EDUCATION

Bachelor of Science in Computer Engineering
Major in Machine Learning
Polytechnic University of the Philippines – Sta. Mesa, Manila

October 2022 – September 2026(Expected)

WORK EXPERIENCE

Technical Operations Intern
Webtek Inc.

July 2025 – August 2025

- Designed UI/UX mockups for web applications and assisted with network device deployment and troubleshooting using Cisco Packet Tracer
- Utilized Photoshop and technical documentation to support field teams and cross-functional engineering projects

Hardware & Operations Intern
Tech Depot IT Solutions Inc.

August 2024 – September 2024

- Provided technical support including hardware diagnostics, PC assembly/disassembly, and product demonstrations
- Resolved hardware issues through systematic troubleshooting and maintained accurate inventory records

RELEVANT PROJECTS

IskoMate: An Academic Assistant for Real-Time Monitoring of Student Engagement

- Developed the end-to-end system architecture integrating Raspberry Pi edge devices and Tailscale to securely stream live webcam data to a central laptop server, supported by a Python Flask backend.
- Engineered two custom computer vision models for distinct "classroom" and "solo" modes, utilizing real-world video datasets collected from live class sessions to optimize sub-second behavior classification.
- Built an interactive Flutter mobile application for session tracking, integrating dynamic zoom-in and zoom-out features for timeline graphs to deliver actionable, real-time participation metrics.

LaTech: Weekly AI & Tech Automation

- Architected a fully automated backend pipeline using Make, the Gemini API, and the Supabase REST API to systematically save gathered weekly data to a Notion workspace and a live website.
- Programmed AI-driven curation workflows to automatically parse, generate, and deliver insights on major AI developments and trending open-source repositories.

QuizCraft: Learning Companion

- Developed an AI-powered study platform using Firebase and the Gemini API to automatically transform uploaded learning materials into customized quizzes and flashcard decks.
- Designed a responsive, SaaS-level academic workspace featuring organized folders to streamline the management and retrieval of user study materials.
- Implemented interactive study tools and automated progress tracking modules to deliver an enhanced and quantifiable user learning experience.

Vrush: AI Video Generation Platform

- Built a full-stack AI video generation platform to successfully transform simple text prompts into fully produced visual content.
- Integrated FFmpeg, Edge TTS, and the Groq API to generate reliable text-to-speech narration and customizable, burned-in captions for dynamic video outputs.

RiceGuard: Rice Plant Health Classification AI

- Trained a custom Support Vector Machine (SVM) model using Scikit-learn and computer vision feature extraction techniques to achieve high accuracy in real-time crop disease detection.
- Deployed a diagnostic web application utilizing Streamlit and the Gemini API to provide users with a highly accessible interface for real-time plant health classification.

TECHNICAL SKILLS

AI & Automation: LLM Integration, Prompt Engineering, API Integration, Workflow Automation, Make, n8n
Languages: Python, TypeScript, JavaScript, Dart
Frontend: Next.js, React, Flutter, HTML, CSS, Tailwind CSS, Vite, Capacitor
Backend & Databases: Node.js, Flask, Supabase, Firebase Firestore, MongoDB, SQLite, SQLCipher
Computer Vision & Machine Learning: OpenCV, MediaPipe, YOLOv8, TFLite, Scikit-learn
Cloud & Infrastructure: Git/GitHub, Vercel, Tailscale, Render, Docker